Demand Analysis with interval-valued sales

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Abstract

Most of existing empirical demand studies assume that market shares are fully observed, at least for some products. The issue of partial observation often raised by empiricists occur when zero sales are recorded for few or many products of interest. However, market sales can still be partially observed when they're recorded in interval values, as it is often seen in the recent and growing industry for apps where downloads/installs data may be disclosed by app providers in intervals (e.g. 100+ downloads, 1000+, 10k+, 100k+ downloads). Interval-valued market sales can also be derived when aggregate market shares are observed. This paper is an attempt to use recent developments in partial identification methods to run empirical demand analysis when observed sales are interval-valued. Simulations show that using the middle of intervals to run point estimation may lead to systematic bias; even if the sign of the bias depends on the DGP. An illustration is done in a conditional logit framework using data on recent casual videogames available on the Steam Platform.

1. Introduction

Existing studies in discrete choice analysis (Berry et al. [1995, 2004], Berry [1994], Berry and Haile [2014], Compiani [2022], Gandhi and Houde [2019], Nevo [2001] have mostly assumed market shares are point-observed so as to recover point-estimates of demand primitives. Some few addressed the issue of censored sales when they were faced with too many zeros in market sales (Gandhi et al. [2023]). Censored sales occur not only when there are zeros in sales data, but also when sales are reported in intervals. Demand estimates in such context is related to censored data regression.

1.1. Background

Censored data issue occurs when the 'True' value of a variable of interest is not known for some observations and truncated with some partial information. When those unknown true values are in the left side (resp. right side) of the distribution and are replaced by a constant value, they are said to be left (right) censored. In case the censored variable is a continuous one, many empirical works have relied on Tobit regression. This method has particularly been popular in the wage analysis as the potential wages for unemployed people are left-censored and truncated to 0. Tobit regression relies on the main assumption that the distribution of the unobvervables is known (and is gaussian). However, this is not a usual assumption in the discrete choice demand analysis where estimating the distribution of the unobservables (here the unobserved product heterogeneities) is of interest for the analyst (Berry et al. [1995]). As an example, Gandhi et al. [2023] add no

distribution assumption and opt for an inference strategy that relies on both moment inequalities and point identification when facing with zeros in market shares for most of their products.

Beside left and right-censored data, data can also be censored when the unknown true value is replaced by an interval where its belongs almost certainly. Interval data regressions cover such cases. In some income analysis, household incomes are often interval-reported in surveys. In such case, one naive solution often employed is to take a 'representative' point within the interval, namely the middle value. However, the underlying assumption is that the middle of interval is representative enough of the whole interval. This is problematic when the censored variable has assymetric distribution around the middle of interval. This is typically true for income and sales whose distributions are known to be skewed. A second simple approach - an extension of the Tobit model to the interval data regression - would be to address this interval data issue by assuming a known distribution of the unobservables and hence run regressions based on likelihood function to point identify the parameters of interest (Bettin and Lucchetti [2012], Stewart [1983]). However, in demand analysis, as mentioned above, it is not convenient to make such assumption as those unobservables are insightfull to researchers. Instead, it is common to assume conditional mean independence over the demand shocks (Berry et al. [1995], Berry and Haile [2014]).

A convenient regression procedure that is consistent with no assumption on the distribution of demand shocks is one based on partial identification. In a context of partial observation of sales, one can still use point identification if the settings are appropriate. For e.g. Gandhi et al. [2023] partially observe sales as they face with many zero market shares. However, they could still use point identification thanks to favorable conditions - even if they have moment inequalities settings. Indeed, since the observed zero market shares are not consistent with a non-zero equilibrium demand, they are able to get lower and upper bounds for the conditional expectation of the inverse demand - the mean utility - leading to an estimation procedure based on moment inequalities. Partial observation of sales inherently leads to partial identification of demand parameters when no point-identification friendly restrictions are added.

Empirical i.o. economists have also used moment inequality restrictions and partial identification for reasons other than partially observed sales. In auctions, the intrinsic values that players have for the biding object are unknown but needed as researchers are interested in the distribution of those private valuations. According to economic theory-based restrictions, the private valuation of a player is lower-bounded by her revealed bid and upper-bounded by the winner's bid. These restrictions are enough to partially identify the distribution of private valuations (Haile and Tamer [2003], Tamer [2010]). Beyond these economic theory-based restrictions, the revealed preference restrictions widely lead to partial identification with many moment inequalities when they're not associated with an assumption of a known distribution for the unobserved heterogeneity.

Interval-valued sales can occur in various cases. In the recent and growing industry of apps,

app providers often disclose their sales data - proxied by the volume of downloads or installs - only partially in intervals. These intervals are in the following format for apps on the Play Store platform: 100+, 1000+, 10k+, 100k+, 1M+, 10M+, etc. As another example, downloads of digital videogames on Steam are accessible through SteamPy and Steam API in interval values. Interval-valued sales can be derived when aggregate sales are recorded by the empiricist while product-specific characteristics are observed. In this case, sales quantity for each product in a given market are derived as lying between 0 and the observed aggregate sales quantity.

As interval-valued demand may lead to moment inequalities without point identification, this paper is an attempt to apply partial identification methods to discrete choice demand estimation when sales are observed in interval values.

1.2. Literature

1.2.1 From early partial identification to recent test inversion from moment inequalities

From old subjects - such as the measurement error leading to bounds in estimating coefficients in linear models (Duncan and Davis [1953], Fréchet [1951], Frisch [1934], Marschak and Andrews [1944]) - to recent issues of identification and inference (Andrews and Shi [2013], Beresteanu and Molinari [2008], Bontemps et al. [2012], Chernozhukov et al. [2007, 2013, 2019], Cox and Shi [2023], Kaido et al. [2019], Manski and Tamer [2002], Pakes et al. [2015]), there are several existing studies in the literature on partial identification. The most recent ones - started with Chernozhukov et al. [2007] - develop a test inversion procedure to get confidence region of the identified set based on moment inequalities (Andrews and Shi [2013], Chernozhukov et al. [2013, 2019], Cox and Shi [2023], Kaido et al. [2019], Pakes et al. [2015]). The challenge those studies usually try to overcome is to develop an inference strategy with less computation cost, more power and less conservative confidence regions. For a comprehensive literature review, we refer to Canay et al. [2023], Molinari [2020], Tamer [2010].

The computation cost-effectiveness sought by some studies (Andrews et al. [2023], Cox and Shi [2023], Kaido et al. [2019], Pakes et al. [2015]) consists in increasing the computation capacity in case there are several parameters. This is convenient in demand analysis when there are various relevant product attributes to consider. Meanwhile, some other studies (Chernozhukov et al. [2019]) have the merit to be robust to an increase in the number of moment inequalities, even above the sample size. As the number of moments increases with the number of instruments, we often have this many moments setting in demand analysis when we use BLP instruments - i.e. other products characteristics. This happens when there are many attributes that the researcher considers as relevant to the consumer choices.

1.2.2 How does a demand model with interval-valued sales fit into partial identification and tests for moment inequalities

This paper is not a first study on demand analysis when sales are partially observed. Gandhi et al. [2023] developed an inference strategy that relies on moment inequalities while still using point identification. Indeed, their moment inequalities derived by the gaps between sales and equilibrium demand become equalities assymptotically. However, this assymptotic point identification cannot apply when all observations have interval-valued market shares as considered in this study. Hence I contribute in censored demand regression model by applying recent developments in partial identification from moment inequalities (Chernozhukov et al. [2019]) to estimate demand when sales are observed in intervals, including sales for the outside good. I show that potential bias could occur when we use the middle of interval as proxy for sales data and run point identification. In our simulations, the bias is quite systematic for a DGP that gives a price distribution with long tail.

2. Model

2.1. Settings

In a market t with J_t app products, the payoff that consumer i gains from getting product j is

$$U_{ijt} = X'_{jt}\theta + \xi_{jt} + \varepsilon_{ijt}$$

where:

- X_{jt}: the observed product characteristics (including endogenous characteristics such as prices) than affect the mean utilities.
- ξ_{jt}: the unobserved product heterogeneity - known to the consumer. Their distribution is assumed unknowned to the analyst.
- ε_{ijt}: unobserved consumer preference. Assume $\{\varepsilon_{ijt}\}_{ijt}$ are assumed iid Type-I EV errors. This implies a conditional logit model and σ_j^{-1} is therefore given by close-form expression: $\sigma_j^{-1}(\mathbf{s_t}) = Log(\frac{s_{jt}}{s_{ot}}) \equiv Log(\frac{S_{jt}}{S_{0t}})$ where S_{lt} refers to sales quantity for good l.

Specifying the distribution of consumer shocks leads to a specification for the demand function $\sigma_j(\delta_t)$ where $\delta_t = (\delta_{1t}, ..., \delta_{J_t t})$ is the vector of mean utilities across products in market t. The inverse $\sigma_j^{-1}(\mathbf{s_t})$ is the mean utility as given by inverting market shares (see Berry et al. [1995] for the inversion). The demand equation to estimate is given by:

$$\sigma_j^{-1}(\mathbf{s_t}) = X'_{jt}\theta + \xi_{jt}$$

2.2. Identification and Estimation method

2.2.1 Market inversion and bounds for mean utilities

We start by considering that the analyst observes the data $W_{jt} = (S_{jt}, S_{jtL}, S_{jtU}, X_{jt}, Z_{jt})_{j,t}$ and denote M_t the value for the exogenous market size.

Assumption 1. *Data Independence (DI)*

$$\textit{Demand shocks } \{\xi_{jt}\}_{j,t} \textit{ are independent across } j, t$$

Assumption 2. *Conditional Logit (CL)*

$$\sigma_j^{-1}(\mathbf{s_t}) = Log(S_{jt}) - Log(S_{ot})$$

Assumption 3. *Partial Observation (PO) of sales*

$$Prob\left(S_{jtL} \leq S_{jt} \leq S_{jtU}\right) = 1 \quad \forall j \in \{0, 1, ..., J_t\}$$

Where bounds S_{jtL}, S_{jtL}; $\forall j \in \{0, 1, ..., J_t\}$; are known by the analyst.

Assumption 4. *Conditional Mean Independence (CMI) of demand shocks*

$$E\left(\xi_{jt}|Z_{jt}\right) = 0$$

Where Z_{jt} is a set of instruments

While assumptions 2, 3, 4 are useful for identification, the assumption 1 is useful for inference. Another implicite assumption relevant for inference is that there are enough variations in the data, i.e. in all partially and point-observed variables related to the demand equation.

Assumption 3 states that realizations of bounds for sales are observed, including bounds for sales quantity of the outside good. Bounds for sales and the bounds for market size are related through: $M_{tb} = S_{otb} + \sum_{l \neq 0} S_{ltb}$; $b \in \{L, U\}$. However, if we initially assumed bounds for market size are observed and not the bounds for outside good, we could derive the bounds for outside good as $S_{otL} = max(M_{tL} - \sum_{l \neq 0} S_{ltU}, 1)$, $S_{otU} = M_{tU} - \sum_{l \neq 0} S_{ltL}$. In both cases, bounds for market size and outside good are observed or derived by the analyst.

In the followings, I describe a pseudo market share inversion. This is a market share inversion - from Berry et al. [1995] - extended to a setting of interval-valued sales with focus on conditional logit model.

From assumption 3 we observe bounds of sales S_{jtL}, S_{jtU} $\forall j$, including bounds for outside good. We know that $\delta_{jt} = Log(S_{jt}) - Log(S_{ot})$ is strictly increasing in S_{jt} and strictly decreasing in S_{ot}. Hence, the Lower bound

$$\delta_{jtL} \equiv \inf_{\{S_{ot}, S_{1t}, ..., S_{J_t t}\}} Log(S_{jt}) - Log(S_{ot})$$
$$s|t \quad S_{otL} \leq S_{ot} \leq S_{otU}$$
$$S_{tL} \leq S_{lt} \leq S_{tU} \quad \forall l \in \{1, ..., J_t\}$$

is reached at

$$\delta_{jtL} = Log(S_{jtL}) - Log(S_{otU})$$

if $S_{jtL}, S_{otU} > 0$ and $\delta_{jtL} = -\infty$ otherwise.

The upper bound

$$\delta_{jtU} \equiv \sup_{\{S_{ot}, S_{1t}, ..., S_{J_t t}\}} Log(S_{jt}) - Log(S_{ot})$$
$$s|t \quad S_{otL} \leq S_{ot} \leq S_{otU}$$
$$S_{tL} \leq S_{lt} \leq S_{tU} \quad \forall l \in \{1, ..., J_t\}$$

is reached at

$$\delta_{jtU} = Log(S_{jtU}) - Log(S_{otL})$$

if $S_{jtU}, S_{otL} > 0$ and $\delta_{jtU} = \infty$ otherwise.

We need the lower bounds S_{jtL}, S_{otL} of the sales quantities to be strictly positive. Otherwise the mean utilities can not be bounded above and bellow simultanously. Moreover, the sales quantity of the outside good is relevant in shaping the size of the confidence region. Indeed, the range of the interval for the mean utility is given by $Log(S_{jtU}) - Log(S_{jtL}) + Log(S_{otU}) - Log(S_{otL})$; which is the sum of the range for log-sales quantity of own-product and the range for log-sales quantity of outside good.

2.2.2 Now the identified set and confidence region

Under assumptions 2, 3 and 4, the identified set as defined by conditional moment inequalities is given by:

$$\Theta_I = \{\theta : E(\delta_{jtL}|Z_{jt}) \leq E(X_{jt}|Z_{jt})'\theta \leq E(\delta_{jtU}|Z_j)\}$$

Θ_I can be expressed equivalently by unconditional moment restrictions as:

$$\Theta_I = \left\{\theta : E[\eta_b g(Z_{jt})(X'_{jt}\theta - \delta_{jtb})] \leq 0 \quad \forall g \in \mathcal{G}, \forall b \in \{L, U\}\right\}$$

Where $\eta_b = -1$ if $b = L$, $\eta_b = 1$ if $b = U$. $\mathcal{G}$ is a convenient countable set of non negative instrumental function. $\mathcal{G}$ is built as hypercubes' indicator functions. This way of getting instrumental

functions was proposed and shown by Andrews and Shi [2013] as suitable to keep information driven by conditional moment restrictions when transforming them into unconditional ones.

Graphs in figures 2, 3 show two examples of an identified set simulated from a DGP with :

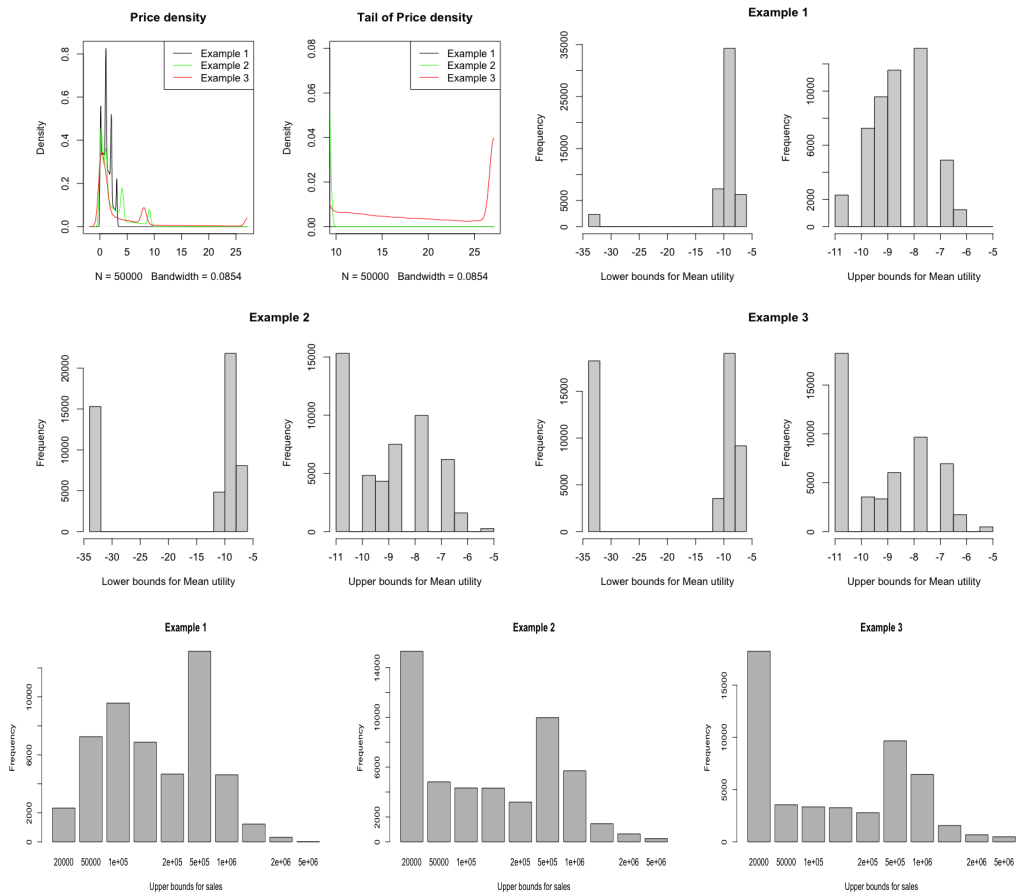
- $J_t = 5$ products
- $X_{jt} = (1, P_{jt})'$
- true parameter $\theta_0 = (\beta_0, \beta_p) = (-7, -1.5)$
- Two instruments $Z_{jt} = (Z_j^1, Z_{jt}^2)$ generated from Bernouilli $Ber(0.5)$
  - Z_j^1 is constant across markets
- With $\lambda_0 = 1$ and $\lambda_p = (-1, 1)'$; the price is generated in 3 different cases according to the level of skewness of its distribution (see figure 1):
  - $P_{jt} = \left|\lambda_0 + Z_{jt}'\lambda_p + \mu_{jt}\right| + 0.1$ for a first example
  - $P_{jt} = \left(\lambda_0 + Z_{jt}'\lambda_p + \mu_{jt}\right)^2 + 0.1$ for a second example.
  - $P_{jt} = \left|\lambda_0 + Z_{jt}'\lambda_p + \mu_{jt}\right|^3 + 0.1$ for a third example.
- $\xi_{jt} = \mu_{jt} + \nu_{jt}$
- $\mu_{jt} = max\left\{-1, min\{\eta_{jt}, 1\}\right\}$; where η_{jt} is a mixture of 5 normal distributions with the following vectors of mixture probabilities $(0.15, 0.3, 0.2, 0.3, 0.15)$, means $(-0.50, -0.25, 0.00, 0.25, 0.50)$ and standard deviations $(1, 1, 1, 1, 1)$.
- ν_{jt} is from $Uniform(-1, 1)$
- The cutoffs used to get intervals for sales are: 1e-5, 2e4, 5e4, 1e5, 1.5e5, 2e5, 5e5, 1e6, 1.5e6, 2e6, 5e6, 1e7, 1.5e7, 2e7, 5e7, 1e8, 1.5e8, 2e8, 5e8, 1e9
- A draw of 100 markets
  - In each draw t, $M_{tL} = M_{tU} = M_t = 10^9$. Hence bounds for sales of outside good are derived: $S_{otL} = max(M_{tL} - \sum_{l \neq 0} S_{ltU}, 0)$ and $S_{otU} = M_{tU} - \sum_{l \neq 0} S_{ltL}$.

As Θ_I is defined by conditional moment restrictions; its set-value represented in figures 2, 3, 4 - by the black-colored shapes - is computed by solving the linear system of inequalities implied by those restrictions. More precisely, it is built by sampling the subsequent linear inverse problem through Markov Chain Monte Carlo (Van den Meersche et al. [2009]).

The goal is to estimate the confidence region for Θ_I from data. Methods developed in recent studies in partial identification generally proceed by inverting a test for moment inequalities:

$$\begin{cases} H_0(\theta): & \max_{v \in \mathcal{V}} E(m_v(\theta, W_{jt})) \leq 0 \\ H_1(\theta): & \max_{v \in \mathcal{V}} E(m_v(\theta, W_{jt})) > 0 \end{cases} \quad (1)$$

Figure 1: Distributions of simulated prices; interval-valued sales and mean utilities



where:

- $v = (b, g)$; $\mathcal{V} = \{L, U\} \times \mathcal{G}$.
- $W_{jt} = (X_{jt}, Y_{jt}, S_{tL}, S_{tU})$ is the observed data.
- $m_v(\theta, W_{jt}) = \eta_b g(Z_{jt})(X'_{jt}\theta - \delta_{jtb})$ is the (unconditional) moment function.

We conclude that a parameter value θ is part of the identified set if the test fails to reject H_0.

The common inference strategy consists in running the test for each of the parameter value in a grid of parameter space, and then gather values that fail to reject H_0 as part of the confidence region. As the moment restrictions originate from lower and upper bounds of the conditional expectation of mean utilities $E(\delta_{jt}|Z_{jt})$, the test defined above at level $\alpha = 0.05$ is actually the intersection of two separate tests of level $\alpha/2 = 0.025$ on : a test for lower bound of $E(\delta_{jt}|Z_{jt})$ and a test for its upper bound. As we put these two together within the same test, the confidence region of size $1 - \alpha = 95\%$ for Θ_I is an inversion of the test 1 at level 2.5%.

Formally, when running test on $H_0(\theta)$, we have:

$$
\begin{aligned}
&Prob(\max_{g\in\mathcal{G}} E(g(Z_{jt})(X'_{jt}\theta - \delta_{jtL})) < 0) \leq \alpha/2 \\
&Prob(\max_{g\in\mathcal{G}} E(g(Z_{jt})(X'_{jt}\theta - \delta_{jtU})) > 0) \leq \alpha/2 \\
\implies &Prob(\max_{(b,g)\in\mathcal{V}} E(\eta_b g(Z_{jt})(X'_{jt}\theta - \delta_{jtL})) > 0) \leq \\
&\quad Prob(\max_{g\in\mathcal{G}} E(g(Z_{jt})(X'_{jt}\theta - \delta_{jtL})) < 0) + Prob(\max_{g\in\mathcal{G}} E(g(Z_{jt})(X'_{jt}\theta - \delta_{jtU})) > 0) \\
&\quad \leq \alpha/2 + \alpha/2 = \alpha
\end{aligned}
$$

The test statistics are in general from the following categories:

- the max statistic: maximum across standardized average moments
- quasi-likelihood ratio statistic
- modified method of moment statistic

I opt in this study for the max statistic from Chernozhukov et al. [2019] (hence CCK test statistic)[1]:

$$T_n(\theta) = \max_{1\leq v\leq k} \frac{\sqrt{n}\bar{m}_v(\theta)}{\hat{\sigma}_{n,v}(\theta)}$$

Where:

- $\bar{m}_v(\theta) = \frac{1}{n}\sum_{j,t} m_v(\theta, W_{jt})$ is the average of the lth moment.
- $\hat{\sigma}_{n,v} = \sqrt{\frac{1}{n}\sum_{j,t}(m_v(\theta, W_{jt}) - \bar{m}_v(\theta))^2}$ is its standard deviation.
- n represents the sample size.

This max-type statistic has some robustness properties as it increases slowly with the number of moment inequalities. Hence, it remains suitable in models with large moment inequalities - relative to sample size n. Moreover, it has some power as it increases weakly with the value of each moment. See Canay et al. [2023], Chernozhukov et al. [2019] for more on the test statistics and on the suitable properties of the max statistic.

Chernozhukov et al. [2019] propose different test methods to compute critical values: a self-Normalised (SN) critical value, which is based on a pivotal statistic (a standard normal); and two bootstrap-based critical values (Multiplier and Empirical bootstrap). They propose three variants for each of these three methods: a one-step variant without moment selection; a two-step variant that discards uninformative moment inequalities that are statistically not binding; and a three-step variant where the authors propose a new selection of "weakly informative inequalities that are potentially binding but do not provide first-order information". In order to conciliate

[1] In a context of data independence, i.e. when the error term ξ_{jt} is independent across observations j, t

the computation effectiveness of the SN method with the power and low conservativeness of the bootstrap methods, they also propose hybrid methods.

In this paper, I am using the two-step hybrid method with multiplier bootstrap critical values in the second step. This choice is motivated by its conciliatory advantage between the computation effectiveness of the SN method in the first step and the power of the bootstrap method in the second step.

3. Simulation Results

For each of our three examples of DGP - distinguished by the price distribution - the identified set and a confidence region from a simulated data sample are computed and shown in figures 2, 3, 4. The confidence intervals that are derived for each coordinate of the parameter θ are presented in table 1 and compared with confidence intervals computed from a 2SLS point estimation when taking the middle of interval. One can see that the confidence intervals computed using point-observed interval middles may not contain the true parameter values. Using a single data sample, the middle of interval leads to estimates of confidence intervals that fail to include the true parameter value for DGP examples 2 and 3.

Table 1: Estimates and 95% Confidence Intervals from examples

	2SLS with middle of interval			CCK (Hybrid)		True Values
	Coef	Lower	Upper	Lower	Upper	(θ_0)
Example 1						
Const	-7.025	-7.271	-6.779	-8.2	-3.4	-7
Price Coef	-1.454	-1.634	-1.273	-5.7	-0.7	-1.5
Example 2						
Const	-7.721	-7.886	-7.555	-8.2	-3.6	-7
Price Coef	-0.790	-0.856	-0.724	-5.6	-0.5	-1.5
Example 3						
Const	-8.118	-8.241	-7.995	-8.4	-6.4	-7
Price Coef	-0.273	-0.294	-0.252	-1.9	-0.2	-1.5

(a) $\delta_{jt} = (Log(S_{jtL} + S_{jtU})) - (Log(S_{otL} + S_{otU}))$

These observations generalize as a middle point bias introduced when running point-estimation based on the interval middle point of sales quantity, and the extend of biases may depend on the DGP. Indeed, when running 500 Monte Carlo simulations in examples 2 and 3, one gets systematic bias in both the price coefficient and the intercept if we use the middle of interval (see table 2). Indeed, when using the middle of the interval, the expected confidence interval of the price

Figure 2: Example 1

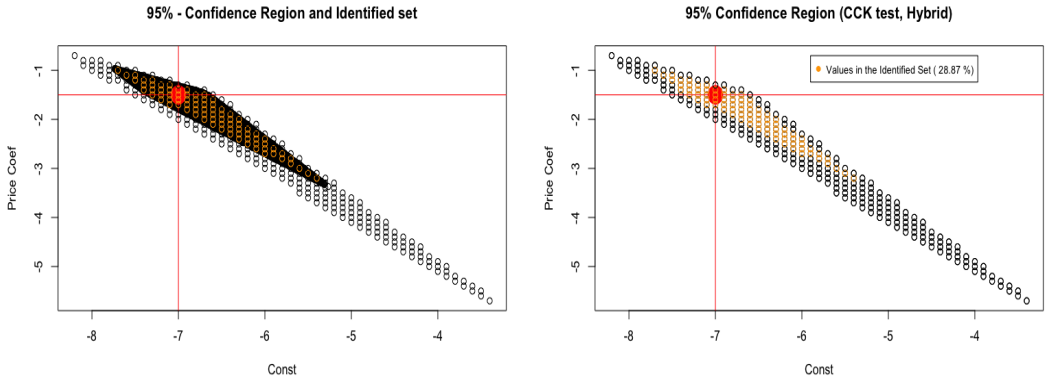


Figure 3: Example 2

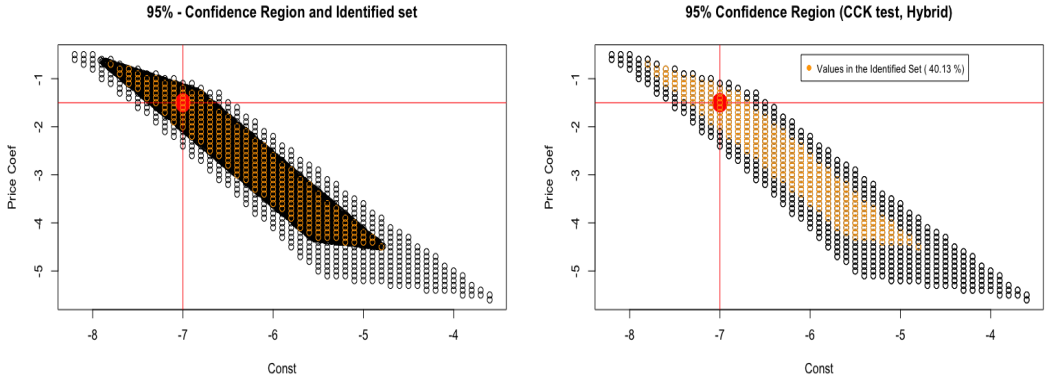


Figure 4: Example 3

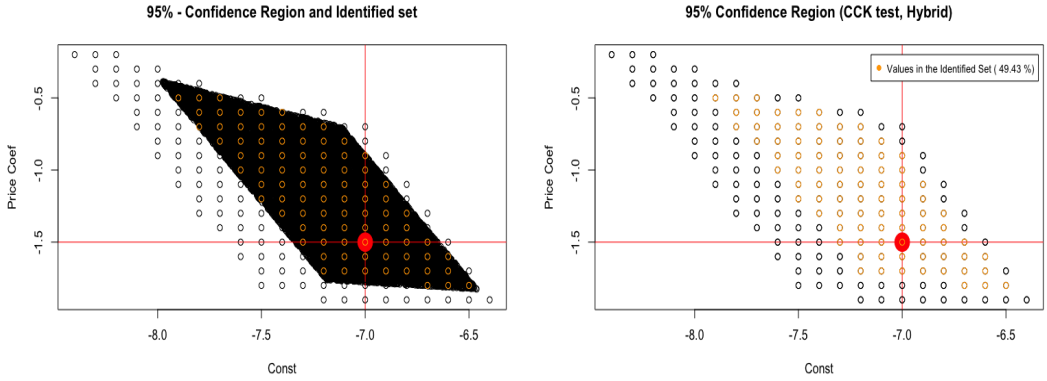


Table 2: 95 %-Confidence Intervals: Averages across 500 simulations

	2SLS with middle of interval			CCK (Hybrid)		True Values
	Coef	Lower	Upper	Lower	Upper	(θ_0)
Example 1						
Const	-6.9	-7.147	-6.653	-8.091	-2.722	-7
Price Coef	-1.539	-1.722	-1.356	-6.197	-0.763	-1.5
% of simulations with true $covering^{(b)}$ in example 1						
Const		90.2%		99.9%		
Price Coef		95.6%				
Example 2						
Const	-7.715	-7.874	-7.556	-8.227	-3.894	-7
Price Coef	-0.774	-0.836	-0.711	-5.25	-0.471	-1.5
% of simulations with true $covering^{(b)}$ in example 2						
Const		0%		99.9%		
Price Coef		0%				
Example 3						
Const	-8.129	-8.262	-7.995	-8.316	-6.057	-7
Price Coef	-0.308	-0.331	-0.285	-2.215	-0.23	-1.5
% of simulations with true $covering^{(b)}$ in example 3						
Const		0%		100%		
Price Coef		0%				

(a) $\delta_{jt} = (Log(S_{jtL} + S_{jtU})) - (Log(S_{otL} + S_{otU}))$

(b) For the 2SLS estimates: % of simulations where the confidence intervals covered θ_0

For the CCK estimates: % of simulations where the confidence region covered Θ_I

coefficient, $[-0.836, -0.711]$ for example 2 and $[-0.331, -0.285]$ for example 3 don't include the true value -1.5 . Likewise, the expected confidence interval of the intercept, $[-7.874, -7.556]$ for example 2 and $[-8.262, -7.995]$ for example 3 do not include the true value -7. Hence, the price coefficient is overestimated while the intercept is underestimated in both DGPs. In the simplified setting of conditional logit that I assume, a DGP with a longer tail of the price distribution leads to a more positive bias in the price coefficient. However, despite being less conservative, the confidence intervals projected from CCK confidence region are not "biased" in a sense that they almost always cover the true parameter value.

4. Industry Data

An industry of interest for implementing the CCK procedure in a demand analysis is the industry for digital apps as app providers often disclose their download numbers in intervals. This is the case for apps that are available on the Google's Playstore platform. I am particularly focused on another case of PC videogames that are on the Steam platform.

Data were collected from Steam API and SteamPy during 11 months on all apps available on Steam: March 2023 and then each month from October 2023 to July 2024. Table 4 summarizes the main observed outcomes such as prices, interval-valued sales and bases of users. Intervals for sales are derived from intervals for cumulative sales (i.e. base of users).

In order to implement the CCK test procedure in demand analysis, I focus on the market for recent apps, i.e. apps that were released between January 2023 and July 2024 included; and that were still available in July 2024. This choice is motivated by the observation periods, lying between March 2023 and july 2024, so that I could only observe price and demand changes within that range of periods. I also include apps that were released in January and February of 2023 as we expect no big changes in demand within that short time period for most of products. I then exclude apps that are not tagged as belonging to following popular genres: action, adventure, role-playing-game, simulation, sports,racing and education. The number of remaining apps in the considered market vary from 3417 in october 2023 to 7070 in july 2024. From those selected apps, 90 are from action-adventure genre and reached at least once the threshold of 20K users between 03-2023 and 07-2024. Those 90 apps are the substitutable products that I am considering in the consumer's choice set. The other apps that are not from action-adventure genre or never reached 20K users are pooled in the outside option. This focus on the 90 popular apps is driven by the quest for enough variability in (interval-valued) sales and user bases. Figure 7 is indicative of the much concentrated distribution of (interval-valued) base of users when we consider all apps available on Steam: As of October 2023, more than half of all apps had same interval of 0-20000. Moreover, by focusing on a quite specific group of products (recent, casual videogames from action-adventure genre), the demand analysis doesn't require to consider many product attributes. I am assuming that the selected 90 apps are differentiated by two observed attributes - price and online/multiplayer mode status - and their unobserved quality.

A summary of product attributes considered in this analysis is provided in table 5. The price (P_{jt}) is an endogenous regressor in the demand equation. Exogenous variations come from the multiplayer/online mode status[2] (MO_{jt}) and two excluded instruments for the price: the number - in log - of total apps the firm has on Steam and a dummy variable for whether or not the firm has a single app available on Steam. The online/multiplayer mode status for rival products couldn't

[2]More precisely, this is a tag variable indicating to users that they have access to at least one of the following features if they get the product: multiplayer, person-versus-person (pvp), co-op and online gameplay

be used as instrument because of low variations. The demand equations in this implementation is given by:

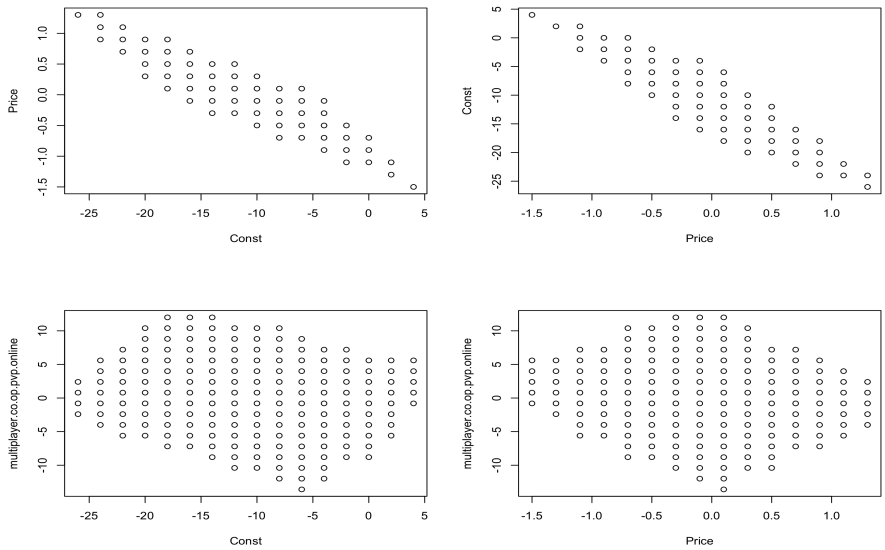
$$\delta_{jt} = \theta_c + \theta_p P_{jt} + \theta_{om} MO_{jt} + \xi_{jt}$$

where $\delta_{jt} = Log(S_{jt}) - Log(S_{ot})$. Sales quantities S_{jt} and S_{ot} are partially observed in intervals. The derived intervals for mean utilities δ_{jt} have bounds that vary as shown in figure 8. There are enough variations in the upper bounds to make inference on demand possible.

Time period are considered as markets. Market units are purchase occasions and the sales quantity of a product is the number of purchase occasions in which consumers chose that product. Hence, market size is the total purchase occasions on Steam. Purchase occasions are assumed independent in their respective demand shocks as stated by assumption 1. As already mentioned, I focus on recent and casual apps from action-adventure genre that could gathered at least 20K units downloaded from Steam platform. Assuming that all purchase occasions led to actual purchase on the platform, the outside option include the other casual and recent apps from big genres that couldn't gather 20K downloads as of July 2024.

5. Estimates from industry data

Figure 5: Projected 95% Confidence Region



I perform the CCK test to build confidence regions for parameters in the demand for recent casual videogames from action-adventure genre; namely the sensitivity to price, the sensitivity to multiplayer/online options and the average taste - for unobserved attributes - of the considered products w.r.t the outside good (i.e. the constant term in θ). In order to visualize its shape, the

Table 3: 95% projected confidence intervals

Coef	Grid		CCK 95%-CI		2SLS 95%-CI$^{(a)}$		
	lower	upper	lower	upper	Coef	lower	upper
Const	-40	10	-26.0	4.0	-8.392	-8.516	-8.269
Price	-2.5	2.5	-1.5	1.3	0.011	-0.0002	0.022
multiplayer/online	-20	20	-13.6	12.0	-0.030	-0.223	0.164

(a) When using the middle of interval)

estimated three-dimensional confidence set is projected in a joined space of two of its coordinates: first, the (intercept, price coefficient) space; and then, the (intercept, multiplayer/online coefficient) space (figure 5). Table 3 shows the projected confidence intervals into the space of each coordinate of the confidence set. I also computed point estimates and confidence intervals from a 2SLS method when using the middle of interval for sales.

Since the zero value for the parameter is included in the confidence region, the information carried by the data are not enough to sustain that consumers have systematically negative sensitivity to price nor positive sensitivity to the multiplayer/online mode. However, we can at least infer from data that if the average taste for unobserved attributes - i.e. the intercept - were high (> -5), consumers would have negative sensitivity to price. Moreover, consumers seems to be not significantly sensitive to the multiplayer/online mode as values of θ_{mo} always cover the zero value for any values of θ_p in the confidence set. Point observation of sales data or additional restrictions in the demand model would be required in order to infer on the exact and unique sensitivity parameters. As shown in previous section from simulations, taking the middle of interval as point-observed values for sales is a restriction that exposes to potential bias, despite leading to quite precise estimates.

6. Conclusion

The purpose of this study was to run demand analysis when market sales are partially observed in interval values. This is a contribution - in demand analysis under partial observation of sales - beside Gandhi et al. [2023] who studied demand when many products have zero market shares. We both rely on an inference strategy based on moment inequalities defined by censored observed sales. However, while their demand is still point identified, I study a partially identified demand since sales are interval-valued for all observations. Hence, I use a recent moment inequality-based inference strategy developed by Chernozhukov et al. [2019] to compute the confidence region of the identified set of parameters. I compare the projected confidence intervals from that confidence region with confidence intervals one would get when assuming point identification of demand using the middle of interval. For that, I run simulations using three simple DGPs that allow the

price to be the sole relevant attribute for consumers and those three DGPs differ in the length of the tail for the price distribution. Simulations show that point-estimates of demand using the middle of interval for sales lead to systematic bias for some DGPs. In particular, among the DGPs used, that bias is deeper for the DGP with longer tail in the price distribution. For this reason, we can rely more on the confidence region obtained by inverting a test for moment inequalities implied by partial identification. I also implement the estimation strategy on demand data on recent and casual PC videogames that belong to the action-adventure genre and that reached at the threshold of 20K users as of July 2024.

A. Appendix A

Figure 6: Price distribution

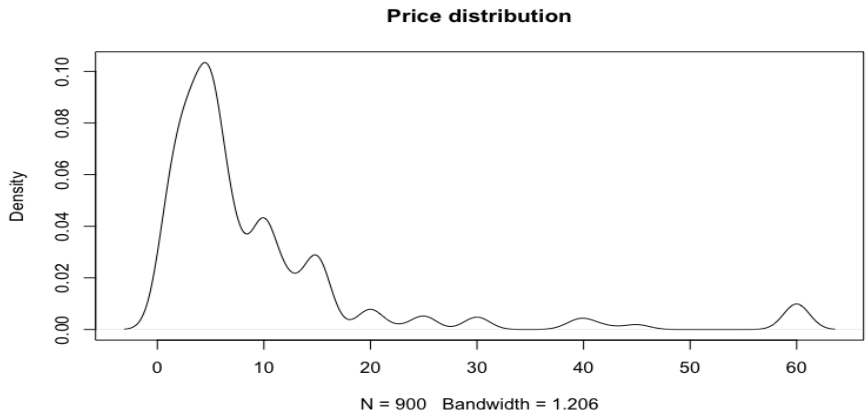


Figure 7: Distribution of interval-valued Base of user for all apps available in Oct 2023

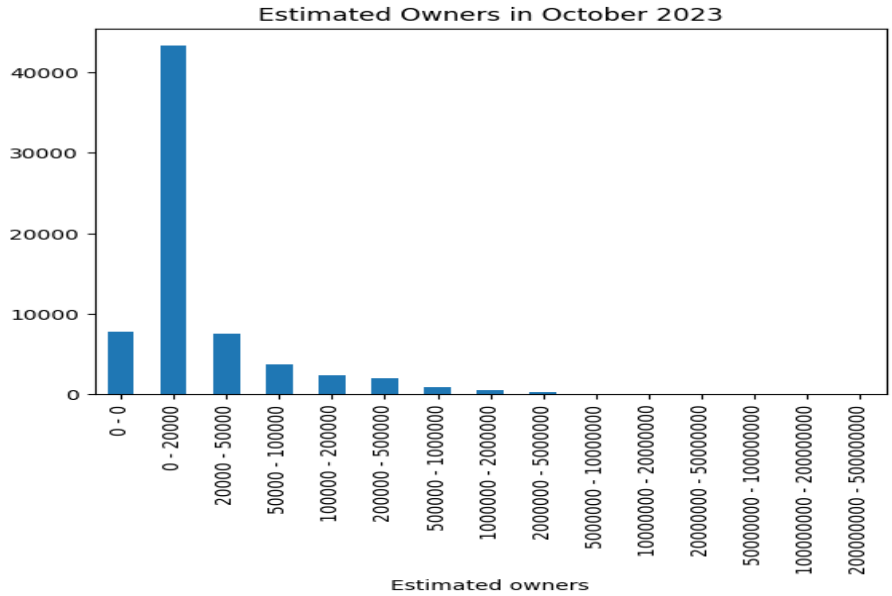


Figure 8: Distribution of Lower and Upper Bounds for δ_{jt}

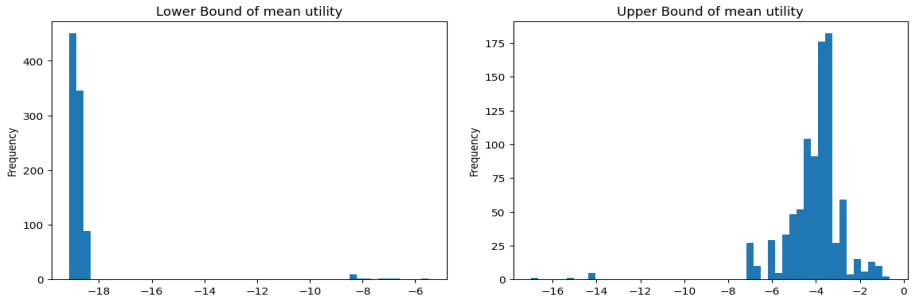


Table 4: Summary of demand data[(a)]

Collect date	10-2023	11-2023	12-2023	01-2024	02-2024	03-2024	04-2024	05-2024	06-2024	07-2024
Nb Apps	3417	3937	4330	4594	4713	5317	5774	6216	6621	7070
Nb in outside option	3327	3847	4240	4504	4623	5227	5684	6126	6531	6980
Nb in inside options	90	90	90	90	90	90	90	90	90	90
Inside options: casual and PTP apps from action-adventure genre, with at least 20K users reached in 07-2024)										
Avg Price	10.371	9.018	10.460	10.344	9.672	6.938	9.813	10.231	9.772	9.913
FTP ratio	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Avg Middle for Base (1e5)	0.603	0.577	0.579	0.587	0.644	0.676	0.676	0.747	0.796	0.817
Avg Lower Bound for Base (1e5)	0.358	0.339	0.341	0.348	0.391	0.413	0.413	0.460	0.490	0.507
Avg Upper Bound for Base (1e5)	0.849	0.816	0.817	0.827	0.898	0.938	0.938	1.033	1.101	1.128
Avg Lower Bound for Sales (1e3)	13.111	0.333	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Avg Upper Bound for Sales (1e3)	73.111	45.778	47.778	48.556	55.000	54.667	52.444	62.000	64.111	63.778
Ratio of zero Lower Bound for Sales	0.833	0.989	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Outside option (apps from other major genres that do not satisfy criteria for inside option apps)										
Avg Price	8.144	8.096	8.383	7.943	7.981	6.244	7.949	8.078	8.371	8.183
FTP ratio	0.012	0.014	0.014	0.016	0.017	0.017	0.016	0.016	0.017	0.017
Avg Middle Base (1e5)	0.215	0.216	0.219	0.225	0.229	0.229	0.224	0.228	0.227	0.227
Avg Lower Bound for Base (1e5)	0.076	0.078	0.079	0.082	0.085	0.086	0.082	0.086	0.085	0.085
Avg Upper Bound for Base (1e5)	0.354	0.354	0.359	0.367	0.372	0.371	0.365	0.371	0.369	0.369
Avg Lower Bound for Sales (1e3)	7.072	1.175	0.330	0.180	0.344	0.631	0.167	0.424	0.214	0.202
Avg Upper Bound for Sales (1e3)	35.206	28.869	28.811	29.203	29.161	29.585	28.561	29.483	28.898	28.910
Ratio of zero Lower Bound for Sales	0.919	0.988	0.994	0.996	0.996	0.990	0.999	0.993	0.994	0.996

(a) focus on recent apps (released since Q1-2023) with positive number of users and from major genres

Table 5: Summary statistics

Variables	count	mean	std	min	25%	50%	75%	max
Lower Bound for Sales (1e5)	900	0.01345	0.177	1	1	1	1	4.8
Upper Bound for Sales	900	0.56722	0.79512	1	0.3	0.3	0.5	10
Lower Bound for Sales of Outside good (1e5)	900	41.51	65.5672	8.1	14	15	33	235.3
Upper Bound for Sales of Outside good (1e5)	900	1504.8	304.05041	1110.6	1221.6	1447.25	1806.1	2017.9
Price (USD)	900	9.653	11.706	0.490	2.990	4.990	9.990	59.990
Instruments for price								
multiplayer/co-op/pvp/online	900	0.133	0.340	0	0	0	0	1
Log-Nb Releases	900	1.614	1.661	0	0	0.896	3.091	5.215
Single Release	900	0.344	0.475	0	0	0	1	1

Table 6: Grid of parameter space

	Grid												
Const	-40	-38	-36	-34	-32	-30	-28	-26	-24.0	-22.	-20	-18	-16
	-14	-12	-10	-8	-6	-4	-2	0	2	4	6	8	10
Price Coef	-2.5	-2.3	-2.1	-1.9	-1.7	-1.5	-1.3	-1.1	-0.9	-0.7	-0.5	-0.3	-0.1
	0.1	0.3	0.5	0.7	0.9	1.1	1.3	1.5	1.7	1.9	2.1	2.3	2.5
multiplayer/co-op/	-20	-18.4	-16.8	-15.2	-13.6	-12	-10.4	-8.8	-7.2	-5.6	-4	-2.4	-0.8
online/pvp Coef	0.8	2.4	4	5.6	7.2	8.8	10.4	12	13.6	15.2	16.8	18.4	20

B. Appendix B

Figure 9: Projected 95% C R.

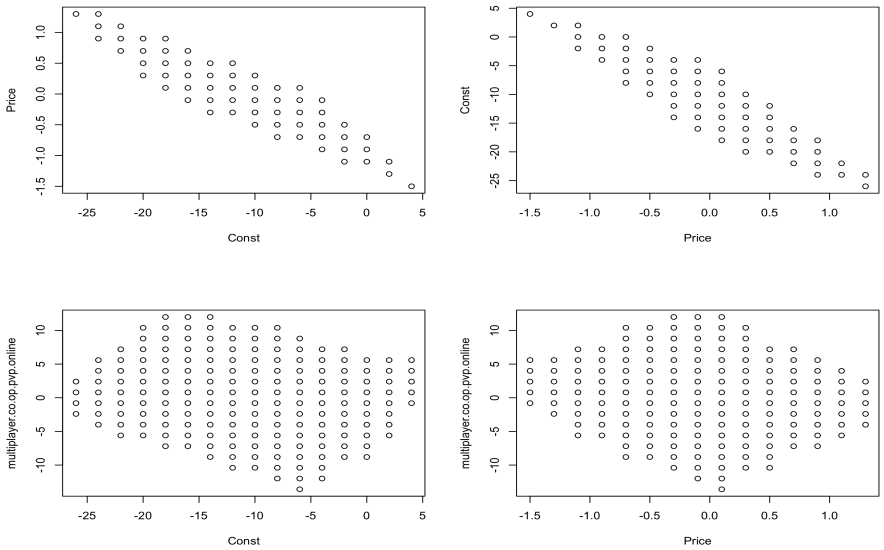


Figure 10: Projected 90% C R.

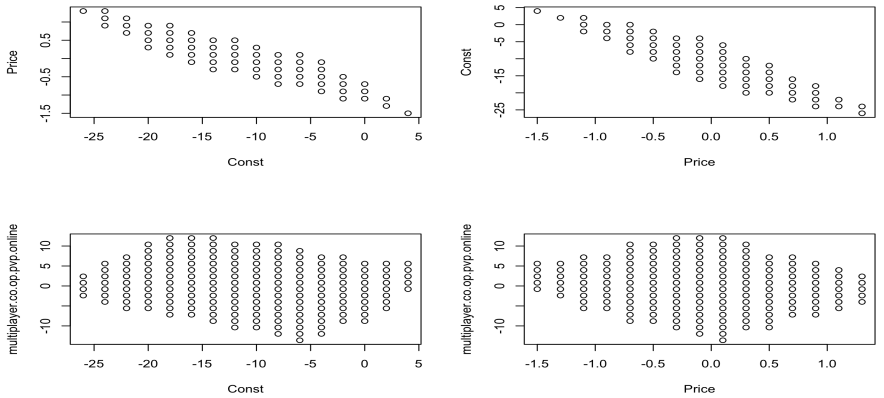
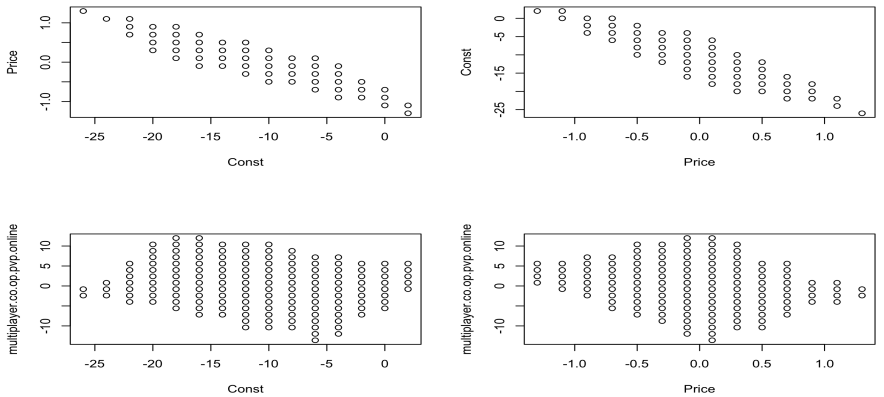


Figure 11: Projected 80% C R.



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